

FETHİYE IRMAK DOĞAN

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OBJECTIVE	Advancing autonomous robots to understand, interact, and collaborate with people
INTERESTS	Human-Robot Interaction, Robot Learning, Explainability, Deep Learning
EDUCATION	<i>Ph.D. in Computer Science, KTH Royal Institute of Technology</i> 01.2018-03.2023 – Thesis: Robots That Understand Natural Language Instructions and Resolve Ambiguities – Thesis Advisor: Assoc. Prof. Iolanda Leite – Thesis Committee: Prof. David Traum (Opponent, USC Institute for Creative Technologies), Assoc. Prof. Jean Oh (Carnegie Mellon University), Assoc. Prof. Luísa Coheur (University of Lisbon) & Dr. Emmanuel Senft (Idiap Research Institute) – 61.0 ECTS credits in PhD-level courses ($\approx$ one academic year; graded pass/fail basis) <i>M.Sc. in Computer Engineering, Middle East Technical University</i> 09.2015-01.2018 – Thesis: <i>Hierarchical Incremental Context Modeling in Robots</i> – Thesis Advisor: Prof. Sinan Kalkan – Graduated with High Honours degree - CGPA: 3.93/4.00 <i>B.Sc. in Computer Engineering, Middle East Technical University</i> 09.2010-06.2015 – Graduated with Honours degree
EMPLOYMENT AND RESEARCH EXPERIENCE	<i>Postdoctoral research associate at University of Cambridge, UK</i> 04.2024-present – Topics: Generating socially appropriate robot actions leveraging LLMs and explainability; Robot-mediated child wellbeing assessment – Projects: ARoEQ (UKRI project), Google GIG project & MICRO (ESRC/UKRI project) – Advisor: Prof. Hatice Gunes, Affective Intelligence and Robotics Laboratory (AFAR) – Outputs: Publication J9, J8, J6, J5, J4, J3, C30, C29, C28, C27, C26, C25, C24, C23, C22, C21, C19, C17, C14 <i>Postdoctoral researcher at KTH Royal Institute of Technology</i> 03.2023-04.2024 – Topics: Continual learning & explainability for robots to assist people’s daily tasks – Project: Partially involved in PerCorSo (WASP project) – Advisor: Prof. Iolanda Leite, Division of Robotics, Perception and Learning – Outputs: Publication J7, C20, C16, C15, C13, C12, W6 <i>Doctoral researcher at KTH (employed for 80% research, 20% teaching)</i> 01.2018-03.2023 – Topics: Generating follow-up clarifications (either semantic or visual) for robots to resolve ambiguities in user instructions – Advisor: Prof. Iolanda Leite, Division of Robotics, Perception and Learning – Outputs: Publication J2, J1, C18, C11, C10, C9, C8, C7, C6, C3, W5, W4, W3, W2 <i>Visiting scholar at Georgia Institute of Technology, USA</i> 11.2021-04.2022 – Topics: Developing a semantically-driven disambiguation method to handle ambiguous user requests with clarifying questions – Advisor: Prof. Sonia Chernova, Robot Autonomy and Interactive Learning (RAIL) Lab – Outputs: Publication C18, C11 <i>Participating in Oxford Machine Learning Summer School</i> 07.2021-08.2021 – Topics: Selected to participate in the highly selective summer school ($\sim$ 15% acceptance rate) for best-in-class training on machine learning and deep learning <i>Participating in Amazon Alexa Prize, KTH Fantom Team</i> 02.2018-08.2018 – Topics: Through a highly competitive process, selected as one of the teams to create a social bot for Amazon Alexa ($\sim$ 4% acceptance rate) – Advisor: Prof. Gabriel Skantze, Division of Speech, Music and Hearing – Outputs: Publication C6, C3 <i>Researcher at Middle East Technical University, Turkey</i> 09.2015-01.2018 – Topics: Incremental context modelling for robots in real-world environments – Project: Context in Robots (TUBITAK project) – Advisor: Prof. Sinan Kalkan, Kovan Robotics Lab – Outputs: Publication C5, C4, C1, W1, T1

HONORS AND AWARDS

- Senior Design Project at Middle East Technical University* 09.2014-06.2015
- **Topics:** 3D animation of fMRI data to visualise the cognitive processes in the brain
 - **Advisor:** Prof. [Fatos Tunay Yarman Vural](#), ImageLab
 - **Outputs:** Publication [C2](#)
- Research intern at **University of Southern Denmark*** 06.2014-09.2014
- **Topics:** A plugin providing a GUI for automated calibration of UR robot arms
 - **Project:** Erasmus internship
 - **Advisor:** Prof. [Norbert Krueger](#), SDU Robotics
- Research intern at Middle East Technical University* 06.2013-09.2013
- **Topics:** Visualising the 2D and 3D representations of the iCub robot’s vision
 - **Advisor:** Prof. [Sinan Kalkan](#), Kovan Robotics Lab
- Part time software developer at **Özgür Yazılım Company*** 02.2013-06.2013
- **Topics:** Taking a role in the development of the Tekir Accounting Program
- **Outstanding Women in Robotics & Auto. Paper Award** Shortlisted 2026
Topic: Shortlisted for Early Career Contribution Award (Publication [C30](#))
Awarded by: IEEE/RSJ Int. Conference on Intelligent Robots and Systems (IROS)
 - **Wiseman Prize**, Dept. of Computer Science & Tech., University of Cambridge 2026
Topic: Recognising exceptional contributions to the Department beyond expected duties
 - **Outstanding Women in Robotics & Automation Paper Award** Finalist 2025
Topic: Top 3 Finalist for Early Career Contribution Award (Publication [C19](#))
Awarded by: IEEE International Conference on Robotics & Automation (ICRA)
 - **KROS Interdisciplinary Research Award in Social HRI** 2025
Topic: “Robot-Led VLM Wellbeing Assessment of Children” (Publication [C17](#))
Awarded by: IEEE Int. Conf. on Robot and Human Interactive Communication (RO-MAN)
 - **IEEE Best Paper Award** and **IEEE Best Student Paper Award** Finalist 2025
Topic: Top 3 Finalist for both award categories (Publication [C17](#))
Awarded by: IEEE Int. Conf. on Robot and Human Interactive Communication (RO-MAN)
 - **Future Digileader**, Digitalize in Stockholm Conference 2025
Topic: The event brings together selected global research leaders and rising stars working to drive digital transformation across academia, industry, government, and civil society
Awarded by: [Digital Futures Research Center](#), KTH Royal Institute of Technology
 - **Seal of Excellence Project Proposal Award** 2025
Topic: Marie Skłodowska-Curie Postdoctoral Fellowship for top applicants above 85% score
Awarded by: European Commission
Project: “Socially Appropriate and Adaptive Robot Behaviour (SAARO)”
 - Special Recognition for Outstanding Reviews 2024
Awarded by: ACM/IEEE International Conference on Human-Robot Interaction
 - Research Associate at **Darwin College** 2024
Topic: Competitive selection process based on research excellence and academic merit
Awarded by: Darwin College, University of Cambridge
 - **DAAD AInet Fellowship** 2022
Topic: Research fellowship for outstanding early-career researchers in AI and robotics
Awarded by: German Academic Exchange Service (DAAD)
 - **RSS Pioneers**, Robotics: Science and Systems 2021
Topic: Pioneers award that gathers the world’s top early-career researchers in robotics
Awarded by: Robotics: Science and Systems (Publication [W5](#))
 - **Honourable mention paper award**, ACM CUI 2019
Topic: “Crowdsourcing a self-evolving dialog graph” (Publication [C6](#))
Awarded by: ACM Conference on Conversational User Interfaces (CUI)
 - High honour certificates (Spring 2014-2015, Fall 2014-2015, Spring 2013-2014, Fall 2013-2014) & honour certificate (Spring 2012-2013), Middle East Technical University

GRANTS AND FUNDS

- Contributor to the **School of Technology Seed Fund Grant** 2024
Awarded by: University of Cambridge
Project: “Can social robots help for mediation and advocacy for students with disabilities?”
Budget: £10.000 GBP
- Travel grant from ACL Annual Conference of NAACL-HLT 2019
Awarded by: North American Chapter of the Association for Computational Linguistics
Budget: \$500 USD
- Travel grant from IEEE ICRA 2018
Awarded by: IEEE International Conference on Robotics and Automation
Budget: \$1137.15 USD
- Travel grant from IEEE/RSJ IROS 2018
Awarded by: IEEE International Conference on Intelligent Robots and Systems
Budget: Up to €500.00 EUR

INVITED TALKS

- Title:** “*Beyond Words: Towards Context-Aware and Adaptive Human-Robot Interaction*”
– **Keynote speaker on Nonverbal Cues for Human-Robot Cooperative Intelligence Workshop**, IEEE/RSJ IROS 2026
- Title:** “*Shaping Robot Behaviour through Explanations and Expectations in HRI*”
– **Keynote speaker on Robo-Identity Workshop**, IEEE RO-MAN 2025
- Title:** “*Autonomous and Explainable Robots in Human Environments*”
– **Keynote speaker on CHIA Early Career Conference** 2025
– Seminar talk at the **Designing Intelligence Lab**, Delft University of Technology 2025
– **Interaction Division Colloquium**, Utrecht University 2025
– **Science seminar**, Darwin College, University of Cambridge 2025
- Title:** “*Robots That Understand Natural Language Instructions and Resolve Ambiguities*”
– **Google DeepMind Research Ready Program**, University of Cambridge 2024
– **Talking Robotics**, a series of seminars about Robotics and AI 2021
– **Oxford Machine Learning Summer School**, Unconference Track 2021
– Seminar talk at **RAIL Research Lab**, Georgia Institute of Technology 2021
– Seminar talk at **Image Lab**, Middle East Technical University 2021

SCIENTIFIC CONTRIBUTIONS

Journal Articles

- J9 **F. I. Doğan***, Y. Lou*, A. Markelius, E. Geijer-Simpson, G. Gredebäck, T. J. Ford, G. Castellano, H. Gunes†, G. Warner†, and J. L. Gibson†. ‘**Designing Social Robots for Inclusive Child Wellbeing Assessment: Insights from Communities Supporting Developmental Language Disorder and Forced Migration**’. (under review for *ACM Transactions on Human-Robot Interaction*)
- J8 M. Nigro, H. Gunes, M. Spitale†, **F. I. Doğan†**. ‘**HiBRIDGE: A Hierarchical Bayesian Neural Network Framework for Interpretable Dialogue Management in Group-Robot Interaction**’. (under review for *Robotics and Autonomous Systems (RAS)*)
- J7 E. Yadollahi*, **F. I. Doğan***, Y. Zhang, B. Nogueira, T. Guerreiro, S. Levy-Tzedek, and I. Leite, ‘**Expectations, Explanations, and Embodiment: Attempts at Robot Failure Recovery**’. (under review for *Int. Journal of Human-Computer Studies*)
- J6 N. Churamani*, **F. I. Doğan***, S. Checker, Y. Malladi, H. L. Chiang, and H. Gunes ‘**More Robots, Better Manners? Federated Continual Learning and A Multi-robot, Multi-view Dataset**’. (under review for *Advanced Robotics Research*)
- J5 A. Markelius, **F. I. Doğan**, J. Bailey, J. I. Gibson, and H. Gunes, ‘**Multimodal Behavioural Evaluation of Social Robot Interaction with Disabled Students: A Disability-Stratified Approach**’. (under review for *ACM THRI*)

* Equal First Authorship

† Equal Senior Authorship

- J4 S. Afroogh, ..., **F. I. Doğan**, ..., J. Zhao ‘Beyond Explainable AI (XAI): An Overdue Paradigm Shift and Post-XAI Research Directions’. (under review for *Nature*)
- J3 W. Tang, **F. I. Dogan**, L. Qing, H. Gunes, ‘AsyReC: A Multimodal Graph-based Framework for Spatio-Temporal Asymmetric Dyadic Relationship Classification’, *IEEE Transactions on Circuits and Systems for Video Technology*, 2025, Volume 36, pp. 3693-3708.
- J2 **F. I. Doğan**, G. I. Melsión, and I. Leite, ‘Leveraging Explainability for Understanding Object Descriptions in Ambiguous 3D Environments’, *Frontiers in Robotics and AI*, Volume 9, pp. 937772, 2023.
- J1 **F. I. Doğan**, S. Gillet, E. J. Carter, and I. Leite, ‘The Impact of Adding Perspective-Taking to Spatial Referencing during Human-Robot Interaction’, *Robotics and Autonomous Systems (RAS)*, Volume 134, pp. 103654, 2020.

Refereed Conference Publications

- C30 R. Karpinski*, **F. I. Doğan***, N. Churamani, Y. Luo, M.A. Graaf, D. Dell’Anna, and H. Gunes, ‘Mind the Context: Continual Learning of Socially Appropriate Robot Actions via Environmental–Social Disentanglement’, *Proceedings of IEEE/RSJ Int. Conference on Intelligent Robots and Systems (IROS)*, 2026. (Accepted)
- C29 E. Ozbey*, **F. I. Doğan***, J. Huang*, and H. Gunes, ‘StARS: Socially Appropriate Robot Actions via a Recommender System-Driven Approach’, *Proceedings of IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2026. (Accepted)
- C28 S. Chiang*, T. Brennan*, **F. I. Doğan**, J. Cheong, and H. Gunes, ‘FAIR_XAI: Improving Multimodal Foundation Model Fairness via Explainability for Wellbeing Assessment’, *Proceedings of ACM International Conference on Multimodal Interaction (ICMI)*, 2026. (Long paper, Accepted)
- C27 A. Markelius*, **F. I. Doğan***, J. Bailey, and H. Gunes, ‘Evaluating Human and LLM-Generated Thematic Analysis in HRI for Vulnerable Populations: A Comparative and Ethical Analysis’, *Proceedings of IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2026. (Accepted)
- C26 A. Ashok, F. Babel, P. Holthaus, R. Khot, K. Bransky Kelly, **F. I. Doğan**, K. Berns, S. Rossi, M. Lee, and G. Laban, ‘Why did My Robot Just Change Personality? Prompting Guidelines for a Grounded Robot Persona in LLM-Based HRI’, *Proceedings of IEEE RO-MAN*, 2026. (Accepted)
- C25 J. Huang, **F. I. Doğan**, and H. Gunes, ‘Reimagining Social Robots as Recommender Systems: Foundations, Framework, and Applications’, *Proceedings of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 406–416, 2026.
- C24 A. Markelius, **F. I. Doğan**, J. Bailey, G. Laban, J. I. Gibson, and H. Gunes, ‘Social Robotics for Disabled Students: An Empirical Investigation of Embodiment, Roles and Interaction’, *Proceedings of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 265–274, 2026.
- C23 **F. I. Doğan***, A. Markelius*, and H. Gunes, ‘Designing Social Robots with Ethical, User-Adaptive Explainability in the Era of Foundation Models’, *Companion of ACM/IEEE Int. Conference on Human-Robot Interaction (HRI)*, pp. 536–541, 2026.
- C22 J. Borgstedt, R. Wullenkord, **F. I. Doğan**, A. Markelius, Y. Lou, E. Geijer-Simpson, E. S. Cross, F. Eyssel, T. Ford, J. L. Gibson, H. Gunes, G. Warner, and G. Castellano, ‘Advancing Child Wellbeing Assessment with AI and Robotics across At-Risk Populations’, *Companion of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 145–149, 2026.
- C21 J. Cheong*, **F. I. Doğan***, A. Markelius*, E. S. Cross, F. Eyssel, G. Castellano, and H. Gunes, ‘Equitable Robotics for Wellbeing (EqRoW)’, *Companion of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 1381–1383, 2026.

* Equal First Authorship

- C20 A. Deichler, J. O'Regan, **F. I. Doğan**, A. Klezovich, Lubos Marcinek, I. Leite, and J. Beskow, 'MM-Conv: A Multimodal Dataset and Benchmark for Context-Aware Grounding in 3D Dialogue', *International Conference on Language Resources and Evaluation (LREC)*, pp. 9240-9253, 2026.
- C19 **F. I. Doğan**, U. Ozturk, G. Cinar, and H. Gunes, 'GRACE: Generating Socially Appropriate Robot Actions Leveraging LLMs and Human Explanations', *Proceedings of IEEE Int. Conf. on Robotics and Automation (ICRA)*, pp. 4330-4336, 2025.
- C18 **F. I. Doğan**, M. Patel, W. Liu, I. Leite, and S. Chernova, 'A Model-Agnostic Approach for Semantically Driven Disambiguation in Human-Robot Interaction', *Proceedings of IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, pp. 649-656, 2025.
- C17 N. I. Abbasi*, **F. I. Doğan***, G. Laban*, J. Anderson, T. Ford, P. B. Jones, H. Gunes, 'Robot-Led Vision Language Model Wellbeing Assessment of Children', *Proceedings of IEEE RO-MAN*, pp. 59-64, 2025.
- C16 A. Leszczynski, S. Gillet, I. Leite, and **F. I. Doğan**, 'BT-ACTION: A Test-Driven Approach for Modular Understanding of User Instruction Leveraging Behaviour Trees and LLMs', *Proceedings of IEEE RO-MAN*, pp. 878-884, 2025.
- C15 E. Bartoli, **F. I. Doğan**, and I. Leite 'STREAK: Streaming Network for Continual Learning of Object Relocations under Household Context Drifts', *Proceedings of IEEE Int. Conf. on Robot and Human Interactive Comm. (RO-MAN)*, pp. 1550-1557, 2025.
- C14 E. Yadollahi, **F. I. Doğan**, M. Romeo, D. Kontogiorgos, P. Qian, and Y. Zhang, '3rd Workshop on Explainability in Human-Robot Collaboration: Real-World Concerns', *Proceedings of ACM/IEEE HRI*, pp. 1994-1996, 2025.
- C13 G. Hadjiantonis, S. Gillet, M. Vázquez, I. Leite, and **F. I. Doğan**, 'Let's move on: Topic Change in Robot-Facilitated Group Discussions', *Proceedings of IEEE Int. Conf. on Robot and Human Interactive Comm. (RO-MAN)*, pp. 2087-2094, 2024.
- C12 E. Yadollahi, M. Romeo, **F. I. Doğan**, W. Johal, M. D. Graaf, S. Levy-Tzedek and I. Leite, 'Explainability for Human-Robot Collaboration,' *Companion of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 1364-1366, 2024.
- C11 M. Pattel*, **F. I. Doğan***, Z. Zeng, K. Baraka, and S. Chernova, 'Semantic Scene Understanding for Human-Robot Interaction', *Companion of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 941 - 943, 2023.
- C10 **F. I. Doğan**, I. Torre, and I. Leite, 'Asking Follow-Up Clarifications to Resolve Ambiguities in Human-Robot Conversation', *Proceedings of ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 461-469, 2022.
- C9 M. Iovino, **F. I. Doğan**, I. Leite, and C. Smith, 'Interactive Disambiguation for Behavior Tree Execution', *Proceedings of IEEE International Conference on Humanoid Robots (Humanoids)*, pp. 82-89, 2022.
- C8 A Panesar, **F. I. Doğan**, and I. Leite, 'Improving Visual Question Answering by Leveraging Depth and Adapting Explainability', *Proceedings of IEEE Int. Conf. on Robot and Human Interactive Communication (RO-MAN)*, pp. 252-259, 2022.
- C7 **F. I. Doğan**, S. Kalkan, and I. Leite, 'Learning to Generate Unambiguous Spatial Referring Expressions for Real-World Environments', *Proceedings of IEEE/RSJ International Conf. on Intelligent Robots and Systems (IROS)*, pp. 4992-4999, 2019.
- C6 P. Jonell, P. Fallgren, **F. I. Doğan**, J. Lopes, U. Wennberg, and G. Skantze, 'Crowdsourcing a self-evolving dialog graph', *Proceedings of the ACM International Conference on Conversational User Interfaces (CUI)*, pp. 1-8, 2019.
- C5 **F. I. Doğan***, İ. Bozcan*, M. Celik, and S. Kalkan, 'Cinet: A learning based approach to incremental context modeling in robots', *Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4641-4646, 2018.

* Equal First Authorship

- C4 **F. I. Doğan**, H. Çelikkanat, and S. Kalkan, ‘A Deep Incremental Boltzmann Machine for Modeling Context in Robots’, *Proceedings of IEEE ICRA*, pp. 2411-2416, 2018.
- C3 P. Jonell, M. Bystedt, **F. I. Doğan**, P. Fallgren, J. Ivarsson, M. Slukova, U. Wennberg, J. Lopes, J. Boye, and G. Skantze, ‘Fantom: A crowdsourced social chatbot using an evolving dialog graph’, *Proceedings of Alexa Prize SocialBot Grand Challenge*, 2018.
- C2 O. Yıldız, **F. I. Doğan**, İ. Öztekin, E. Mızrak, and F. T. Y. Vural, ‘A robust normalization method for fMRI data for brain decoding’, *Proceedings of IEEE Signal Processing and Communication Application Conference (SIU)*, pp. 2269-2272, 2016.
- C1 **F. I. Doğan**, S. Kalkan, ‘The Hierarchical Nature of Context’, *Turkey Robotics Conference (ToRK)*, 2016. (written in Turkish)

Refereed Workshop Contributions

- W6 E. Bartoli, **F. I. Doğan**, and I. Leite, ‘Contextualized Knowledge Graph Embeddings for Activity Prediction in Service Robotics’, *Workshop on SSU for HRI*, HRI 2023.
- W5 **F. I. Doğan**, ‘Social Robots That Understand Natural Language Instructions and Resolve Ambiguities’, *RSS Pioneers Workshop*, RSS 2021.
- W4 I. Torre*, **F. I. Doğan***, and D. Kontogiorgos*, ‘Voice, Embodiment, and Autonomy as Identity Affordances’, *Workshop on Robo-Identity*, HRI 2021.
- W3 **F. I. Doğan**, and I. Leite, ‘Open Challenges on Generating Referring Expressions for Human-Robot Interaction’, *Workshop on NLG for HRI*, INLG 2020.
- W2 **F. I. Doğan**, S. Kalkan, and I. Leite, ‘Learning to Generate Unambiguous Spatial Referring Expressions for Real-World Environments’, *SpLU-RoboNLP Workshop*, Annual Conference of the North American Chapter of ACL (NAACL-HLT), 2019.
- W1 **F. I. Doğan**, H. Çelikkanat, I. Bozcan, and S. Kalkan, ‘Learning to Increment A Contextual Model’, *Workshop on Continual Learning*, NeurIPS 2018.

POSTERS AND DEMOS

- AI-CARING Symposium 2022
- Invited demo on SIGDIAL Conference 2019
- SpLU-RoboNLP Workshop at the NAACL-HLT Conference 2019
- SoRos Workshop 2018 and 2019
- Amazon Alexa Prize Summit 2018

TEACHING

- Guest Lectures in the Affective AI Course, University of Cambridge* 2024, 2025 & 2026
- **Provided Lecture:** “Explainability for Social/Affective AI”
 - **Course:** L344 Affective Artificial Intelligence
 - **Course level:** MPhil in Advanced CS and Part III (9-month taught master’s course)
 - **# of students registered:** around 25 students per term
- Teaching Assistant at Interaction Design, University of Cambridge* 04.2024-06.2024
- **Role:** Group project supervision for user-centred interaction design
 - **Course:** Interaction Design
 - **Course level:** Part IA CST (first year undergraduate in Computer Science)
 - **# of students registered:** more than 100
- Lecturer at Social Robotics Course, KTH* 10.2023-01.2024
- **Provided Lectures:** “Perception of human social signals”, “Dialogues, verbal/non verbal communication”, “Social Learning for robots” & “Explainability & Theory of Mind in HRI”
 - **Role:** Lectures, robotics tutorial, and group project supervision for social robot interaction
 - **Course:** DD2413 Social Robotics
 - **Course level:** MS students in Computer Science and Engineering programs
 - **# of students registered:** 34 students
- Teaching Assistant at Machine Learning Course, KTH* 01.2019-03.2022
- **Role:** Lab assignment support and examination on traditional ML methods (Bayes Classifiers and Boosting, Support Vector Machines, Decision Trees)
 - **Course:** DD2421 Machine Learning, 2019-2020, 2020-2021, 2021-2022 (offered twice per year)
 - **Course level:** MS students in Computer Science and Engineering programs
 - **# of students registered:** more than 300 students per semester

Teaching Assistant at C programming language course, METU 02.2014-06.2014

- **Role:** Lab examination support on C programming language course
- **Course:** [CENG140 C programming](#)
- **Course level:** BS students in Engineering programs
- **# of students registered:** more than 300 students

SUPERVISION

PhD Students

- Massimiliano Nigro (visiting student at Cambridge) 2025-present
Program & Home Institution: PhD in Computer Science, Politecnico di Milano
Project: Robot-led group discussion, context and referee prediction
Role: Project co-supervisor (Publication [J8](#), equal senior author)
- Wang Tang (visiting student at Cambridge) 2024-2025
Program & Home Institution: PhD in Computer Science, Sichuan University
Project: Asymmetric dyadic relationship classification
Role: Project co-supervisor (Publication [J3](#))
- Ermanno Bartoli (PhD student at KTH) 2023-2024
Program & Institution: PhD in Computer Science, KTH Royal Institute of Tech.
Project: Continual learning for dynamic object location changes in household setups
Role: Project co-supervisor (Publications [C15](#) and [W6](#))
- Anna Deichler (PhD student at KTH) 2023-2024
Program & Institution: PhD in Computer Science, KTH Royal Institute of Tech.
Project: Multimodal dataset gathering for referential referring expressions in VR
Role: Project co-mentor (Publication [C20](#))

MS Students

- Casper Adolfse (visiting student at Cambridge) 2026-present
Program & Home Institution: MS in Data Science & AI , Eindhoven University of Tech.
Project: Generating animations for simulated robot faces
Role: Project co-supervisor
- Bob Zhang (MPhil student at Cambridge) 2025-2026
Program & Institution: MPhil in Advanced Computer Science, University of Cambridge
Thesis: Mitigating movie recommendation biases for recommendation systems
Role: MS thesis co-supervisor
- Rafal Karpiński (visiting student at Cambridge) 2024-2026
Program & Home Institution: MS in Computer Science, Utrecht University
Thesis: Informing Socially Appropriate Robot Behaviour under Data Distribution Shifts using Continual Learning
Role: MS thesis supervisor (Publication [C30](#))
- Jeshur Joshua (visiting student at Cambridge) 2024-2025
Program & Home Institution: MS in Computer Science, Vellore Institute of Technology
Project: Unsupervised clustering of user nonverbal cues in journaling sessions
Role: Project co-supervisor
- Alexander Leszczynski (MS student at KTH) 2024-2025
Program & Institution: MS in Computer Science, KTH Royal Institute of Technology
Thesis: Leveraging LLMs and Behavior Trees for Understanding User Instructions
Role: MS thesis supervisor (Publication [C16](#), senior author)
- Aiman Shenawa (MS student at KTH) 2024-2025
Program & Institution: MS in Computer Science, KTH Royal Institute of Technology
Thesis: Task specific evaluation of LLMs: A study for human-robot interaction
Role: MS thesis supervisor
- Georgios Hadjiantonis (MS student at KTH) 2023-2024
Program & Institution: MS in Computer Science, KTH Royal Institute of Technology
Thesis: ML for topic change in robot-moderated discussions using non-verbal features
Role: MS thesis supervisor (Publication [C13](#), senior author)
- Amrita Panesar (MS student at KTH) 2021-2022
Program & Institution: MS in Computer Science, KTH Royal Institute of Technology
Thesis: Improving visual question answering with depth and adapting explainability
Role: MS thesis supervisor (Publication [C8](#))

- Aswin Gururaj Prakash (MS student at Georgia Tech) 2022
Program & Institution: MS in Computer Science, Georgia Institute of Technology
Project: Fusing semantic object understanding to the robot’s semantic mapping
Role: Project co-supervisor
- Jiaming Huang (MS student at KTH) 2022
Program & Institution: MS in Computer Science, KTH Royal Institute of Technology
Project: Concise unambiguous referring expression generation to handle uncertain requests
Role: Robotics project course supervisor
- Shipra Jain (MS student at KTH) 2019
Program & Institution: MS in Computer Science, KTH Royal Institute of Technology
Project: Referring Expression generation for human-robot interaction
Role: Robotics project course supervisor

BS Students

- Berk Sezer (visiting student at Cambridge) 2026-present
Program & Home Institution: BS in Computer Eng., Middle East Technical University
Project: Socially Appropriate Robot Action Generation in Office Environments
Role: Project supervisor
- Mehmet Alp Ozaydin (visiting student at Cambridge) 2026-present
Program & Home Institution: BS in Computer Eng., Middle East Technical University
Project: Mitigating Associational Biases in Compound Generative Systems
Role: Project supervisor
- Erencem Ozbey (visiting student at Cambridge) 2025-2026
Program & Home Institution: BS in Computer Engineering, Bogazici University
Project: Socially Appropriate Robot Action Generation Using Recommender Systems
Role: Project co-supervisor (Publication C29)
- Rahma Elsheikh (visiting student at Cambridge) 2024-2025
Program & Home Institution: BS in Math & Computer Science, Princeton University
Thesis: Approaching Equalized Odds by Actively Forgetting in Deep-Learning Models
Role: Senior thesis supervisor
- Kajal Patel (visiting student at Cambridge) 2025
Program & Home Institution: BS in CS, University of Illinois at Urbana-Champaign
Project: Explanation generation for inter-model communication of generative models
Role: Project supervisor (Preprint on [ArXiv](#))
- Yuval Weiss (BS student at Cambridge) 2025
Program & Institution: BS in Computer Science, University of Cambridge
Project: Bias detection for inter-model communication of generative models
Role: Affective AI course project supervisor (Preprint on [ArXiv](#))
- Yasaswi Malladi (BS student at Cambridge) 2025
Program & Institution: BS in Department of Engineering, University of Cambridge
Project: Generating animation of MannersDB+ scenes in Unity
Role: Project supervisor
- Lara Horne (BS student at Cambridge) 2025
Program & Institution: BS in Computer Science, University of Cambridge
Project: Human activity detection and generating explanations
Role: Part II project supervisor
- Sujith Sai (visiting student at Cambridge) 2025
Program & Home Institution: BS in Chemical Eng., Rourkela National Institute of Tech.
Project: ML methods for children’s wellbeing assessment in robot-led discussions
Role: Project co-supervisor
- Zeynep Altundal (visiting student at Cambridge) 2025
Program & Home Institution: BS in Computer Engineering, Sabanci University
Project: MannersDB+ dataset annotation examination and correction
Role: Project supervisor
- Umut Ozyurt (visiting student at Cambridge) 2024
Program & Home Institution: BS in Computer Engineering, Middle East Tech. University
Project: Uncertainty detection for social appropriateness of robot actions
Role: Project supervisor (Publication C19)

COMMUNITY SERVICE

- Gizem Cinar (visiting student at Cambridge) 2024
Program & Home Institution: BS in Psychology, Bilkent University
Project: User explanation categorisation of MannersDB+ dataset
Role: Project supervisor (Publication *C19*)
- Yifei Shi (Google DeepMind intern at Cambridge) 2024
Program & Home Institution: BS in Computer Science, King's College London
Project: End-to-end detection of interaction ruptures and Grad-CAM explanations
Role: Project supervisor
- Research Engineers*
- Alex Sleat 2023
Institution: KTH Royal Institute of Technology
Project: MTurk setup for dataset collection using Matterport 3D home scan videos
Role: Project supervisor
- Rasmus Rudling 2021
Institution: KTH Royal Institute of Technology
Project: System deployment (asking follow-up clarifications) to the Pepper robot
Role: Project supervisor
- Shreya Kohli 2020
Institution: KTH Royal Institute of Technology
Project: Deploying Grad-CAM explainability activations to AI Habitat platform
Role: Project supervisor
- Editorial Role*
- **Associate Editor, *IEEE RA-L***, Human-Robot Interaction Track 2025-present
- **Associate Editor, *IEEE ICRA***, Robot Learning Track 2026 & 2027
- Conference Organization Committee & Chairing*
- **Conference Chair, Oxbridge Conference**, University of Oxford 2026
- Chair of Modelling Social Norms for Robot Navigation Session, IEEE/RSJ IROS 2026
- Chair of Reading and Respecting Social Norms in HRI Session, IEEE/RSJ IROS 2026
- Chair of Human-Robot Interaction Session, IEEE ICRA 2025
- Workshop Organisation Committee & Chairing*
- **Human-Scene Interaction (HSI) Workshop and Challenge**, ECCV 2026
- **Equitable Robotics for Wellbeing (EqRoW)**, ACM/IEEE HRI 2026
- **Neuroergonomics and Physiological Computing for HRI**, IEEE RO-MAN 2026
- **Expl. in Human-Robot Collaboration: Real-World Concerns**, ACM/IEEE HRI 2025
- **Robotics and Embodied Intelligence Workshop**, CHIA, University of Cambridge 2025
- **Explainability for Human-Robot Collaboration**, ACM/IEEE HRI 2024
- **Workshop on Semantic Scene Understanding for HRI**, ACM/IEEE HRI 2023
- **Workshop on HRI for Explainable Robotics**, IEEE RO-MAN 2023
- **RSS Pioneers Workshop**, Robotics: Science and Systems (RSS) 2022
- Program Committee*
- European Conference on Artificial Intelligence (ECAI) 2025
- International Conference on Multimodal Interaction (ICMI) 2024 & 2025
- Affective Computing and Intelligent Interaction Conference (ACII) 2024
- Towards Autonomous Robotic Systems Conference (TAROS) 2021
- SpLU-RoboNLP Workshop 2021 & 2023
- Journal Article Referee*
- IEEE Robotics and Automation Letters (RA-L)
- International Journal of Social Robotics (IJSR)
- ACM Transactions on Human-Robot Interaction (THRI)
- Autonomous Robots (AURO)
- Frontiers in Robotics and AI
- User Modeling and User-Adapted Interaction (UMUAI)

Conference Paper Referee

- Robotics: Science and Systems (RSS)
- Conference on Robot Learning (CoRL)
- IEEE International Conference on Robotics and Automation (ICRA)
- ACM/IEEE International Conference on Human-Robot Interaction (HRI)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
- ACM International Conference on Multimedia (ACMMM)
- ACM International Conference on Multimodal Interaction (ICMI)
- IEEE International Conf. on Robot & Human Interactive Communication (RO-MAN)
- IEEE Signal Processing and Communication Application Conference (SIU)
- Turkey Robotics Conference (ToRK)

Student Volunteer

- ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2022
- Int. Conf. on Autonomous Agents and Multiagent Systems (AAMAS) 2018
- Featured in **BBC Click by Spencer Kelly** for his 1000th and final episode 2025
- **Human Machine Interaction Research Showcase**, University of Cambridge 2025
- **Sutton Trust Summer School Lecture**, University of Cambridge 2025
- **Physics at Work Exhibition**, University of Cambridge 2025
- **HE+ Hampshire Lecture**, Trinity Hall, University of Cambridge 2025
- **Trinity Hall Science Open Day**, University of Cambridge 2025
- **Swedish Foundation for Strategic Research (SSF)** 2023
- KTH School of Electrical Engineering and Computer Science Lab Tours 2023
- Atlanta Science Festival, Georgia Tech Science and Engineering Day 2022
- Featured in a documentary on **Sveriges Television (SVT)** 2020
- **Sweden's Minister for Higher Edu. and Research (Matilda Ernkrans)** 2019
- Robots Exhibition at the **Swedish Tekniska Museum** 2019
- Giants event for female and non-binary high school students 2018
- **Live broadcast on national TV (Kanal B)** on image processing and AI 2017
- European Researchers' Night event for high school students 2016

**SELECTED
OUTREACH**

**EXTRA-
CURRICULAR**

- Member of **Women@CL Committee**, University of Cambridge 2025-present
- Member of the **Dept. Research Staff Forum**, University of Cambridge 2024-present
- Theater (played in private theatres for 8 years and 5 plays) 2010-2019
- **Computer Engineers' Association** board member 2016-2018
- **Student Delegate Committee** member at Middle East Technical University 2010-2011